

Carmel Holy Word Secondary School
2024 – 2025 Term 2 Sample Paper

Secondary 3 Computer Literacy
Question Book

Date : 29th May 2025
Time Limit : 30 minutes
Pages : 7
Total mark : 40

Instructions:

1. You should write your name, class, class number and the examination venue in the spaces provided on Page 1 after the examination start announcement is made.
2. There are three sections, A, B and C, in this Paper.
3. **ANSWER ALL QUESTIONS.** Answers to Section A, B and C should be written in the answer sheet provided.

Name	
Class	
Class No.	
Venue	

By Markers

Total	/ 40
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Section A: Multiple Choice Questions (15 marks)

Choose the most suitable answers.

1. What is the output of the following program?

```
print("I don\t know.")
```

- A. I don\t know.
- B. I don"t know.
- C. I don't know.
- D. This program cannot be compiled.

2. Which of the following codes can be compiled successfully?

- A. 4you = Jesus
- B. class = "3E"
- C. _value_3 = 3
- D. my-var = 30

3. What is the output of the following program?

```
i = "2"  
j = int(i + i)  
print(j * 2)
```

- A. 2222
- B. 84
- C. 66
- D. 44

4. Read the following program.

```
A = input()  
B = input()  
print(A + " and " + B + " are best friends.")
```

Input the following text in the program in order:

```
Peter  
John
```

What is the output of the program in the terminal?

- A. Peter and John are best friends.
- B. A and B are best friends.
- C. John and Peter are best friends.
- D. This program cannot be compiled.

5. Read the following program.

```
n = int(input())  
print(str(n * 2) + str(n ** 2))
```

Input the following text in the program:

3

What is the output of the program in the terminal?

- A. 69
 - B. 39
 - C. 36
 - D. 66
6. Which of the following codes **cannot** print text Z2026 in the terminal?
- A. `print("Z" + 2026)`
 - B. `print("Z" + str(20 * 100 + 26))`
 - C. `print("Z" + str("2026"))`
 - D. `print("Z" + str(2026))`
7. Read the following program.

```
A = input()  
B = A + input()  
print(A + B)
```

Input the following text in the program in order:

K
A

What is the output of the program in the terminal?

- A. KAK
- B. KKA
- C. AKA
- D. AKK

8. Read the following program.

```
a = int(input())  
b = int(input())  
print((a % 5) + b)
```

Input the following text in the program in order:

10
20

What is the output of the program in the terminal?

- A. 10
- B. 20
- C. 30
- D. 40

9. What is the output of the following program?

```
num = 150  
if num > 100:  
    num -= 10  
elif num >= 100:  
    num -= 5  
print(num)
```

- A. 150
- B. 145
- C. 140
- D. 135

10. What is the output of the following program?

```
x = 2  
while x < 20:  
    x = x * 2  
print(x)
```

- A. 16
- B. 32
- C. 20
- D. 2

11. What does the following code output?

```
import random
print(random.randint(5, 7))
```

- A. Always prints 6
- B. Always prints 4 or 8
- C. Prints a random integer: 5, 6, or 7
- D. Error because `randint()` cannot be used

12. What library does the following code require to work?

```
sr.Recognizer()
```

- A. gTTS
- B. OpenCV
- C. GoogleCloudTranslate
- D. SpeechRecognition

13. What is the purpose of `cv2.imshow()` do?

- A. Displays a video/image window
- B. Converts image to grayscale
- C. Translates text
- D. Records speech

14. Which of the following is correct about GoogleCloudTranslate?

- A. It converts text to mp3 audio
- B. It needs internet access
- C. It requires playsound to play audio
- D. It can translate text into Chinese

15. Which line captures audio from the user?

- A. `cv2.VideoCapture(0)`
- B. `tts.save("test.mp3")`
- C. `recognizer.listen(source)`
- D. `client.translate("Hi")`

Section B: Short Questions (15 marks)

All answers should be written in the spaces provided in the answer sheet.

1. Complete the following program to open and show an image using OpenCV. (5 marks)

cv2 imread imshow waitKey destroyAllWindows

```
import ____ (i) ____  
img = cv2.____ (ii) ____ ("sample.jpg")  
cv2.____ (iii) ____ ("Preview", img)  
cv2.____ (iv) ____ (0)  
cv2.____ (v) ____ ()
```

2. Complete the code to make the computer say what the user types using TTS. (5 marks)

gtts gTTS save input playsound

```
from gtts import ____ (i) ____  
from playsound import ____ (ii) ____  
text = ____ (iii) ____ ("Say something: ")  
tts = gTTS(text, lang="en")  
tts.____ (iv) ____ ("voice.mp3")  
____ (v) ____ ("voice.mp3")
```

3. Complete the code for translating a user-inputted sentence from Chinese to English using Google Cloud Translate. (5 marks)

Client result input zh en

```
from google.cloud import translate_v2 as translate  
import os  
os.environ["GOOGLE_APPLICATION_CREDENTIALS"] = "translate.json"  
client = translate.____ (i) ____ ()  
text = ____ (ii) ____ ("請輸入中文句子: ")  
____ (iii) ____ = client.translate(text,  
    source_language="____ (iv) ____",  
    target_language="____ (v) ____")  
print(result["translatedText"])
```

Section C: Long Question (10 marks)

1. You are given the following program that performs three separate tasks using Python. Each part contains a bug or missing logic.

Line	Program
1	# Part A
2	import random
3	print("A random number is:")
4	print(random.randint())
5	
6	# Part B
7	from gtts import gTTS
8	tts = gTTS("Good job", lang="en")
9	tts.save()
10	print("Audio saved.")
11	
12	# Part C
13	import cv2
14	cap = cv2.VideoCapture("lesson.mp4")
15	ret, frame = cap.read()
16	cv2.imshow(frame, "Video Frame")
17	cv2.waitKey(0)

- (a) Identify and explain one error or problem in each part. (6 marks)
- (b) Write a complete and correct program to do the following task: (4 marks)
- Ask the user to input a number
 - Multiply the number by 10
 - Speak out the result using gTTS
 - The program must save and play the voice

End of Paper

Carmel Holy Word Secondary School
2nd Term Sample Paper 2024 - 2025
Secondary 3 Computer Literacy
Answer Sheet

/40

Name : _____

Time allowed : 30 minutes

Section A: Multiple Choice Questions (15 marks)

Choose the most suitable answers. Use an **HB pencil** to blacken the circle next to the question numbers below.

	Class	Class No.	
ZIPGRADE.COM	1 (A) (B) (C) (D)	13 (A) (B) (C) (D)	15-Q (2574)
	2 (A) (B) (C) (D)	14 (A) (B) (C) (D)	
	3 (A) (B) (C) (D)	15 (A) (B) (C) (D)	
	4 (A) (B) (C) (D)		
	5 (A) (B) (C) (D)		
	6 (A) (B) (C) (D)		
	7 (A) (B) (C) (D)		
	8 (A) (B) (C) (D)		
	9 (A) (B) (C) (D)		
	10 (A) (B) (C) (D)		
	11 (A) (B) (C) (D)		
	12 (A) (B) (C) (D)		

Section B: Short Questions (15 marks)

All answers should be written in the spaces provided.

1.	(i)	(ii)	(iii)
	(iv)	(v)	
2.	(i)	(ii)	(iii)
	(iv)	(v)	
3.	(i)	(ii)	(iii)
	(iv)	(v)	

Section C: Long Question (10 marks)

1. (a) Part A:

Error: _____

Correction: _____

Part B:

Error: _____

Correction: _____

Part C:

Error: _____

Correction: _____

(b)

End of Paper

2024-2025 Term 1 S3 CMP Examination
Suggested Answer

Section A: @ 1mark

BCABB CCDDDB CBADA

Section B:

1.	(i) cv2	(ii) imread	(iii) imshow
	(iv) waitKey	(v) destroyAllWindows	
2.	(i) gTTS	(ii) playsound	(iii) input
	(iv) save	(v) playsound	
3.	(i) Client	(ii) input	(iii) result
	(iv) zh	(v) en	

Section C:

1. (a) Part A
Error: **random.randint()** requires two arguments (start, end)
Correction: **random.randint(1, 10)** or any two integers
Part B
Error: **tts.save()** is missing filename argument
Correction: **tts.save("output.mp3")**
Part C
Error: **Incorrect argument order in cv2.imshow()**
Correction: **cv2.imshow("Video Frame", frame)**
- (b)

```
from gtts import gTTS
from playsound import playsound

number = int(input("Enter a number: "))
result = number * 10
tts = gTTS("The result is " + str(result), lang="en")
tts.save("result.mp3")
playsound("result.mp3")
```